

PROCESS PLANT DEBOTTLENECKING AND CAPACITY EXPANSION

Using Aspen Plus V14

Case Study: Binary Benzene–Toluene Distillation

Project Information	Details
Software	Aspen Plus V14
Simulation Model	RadFrac equilibrium-stage distillation column
Feed	50 mol% benzene + 50 mol% toluene
Base Throughput	1000 kg/h
Expanded Throughput	1300 kg/h (+30%)
Debottlenecking Variable	Reflux ratio

Prepared as an engineering simulation and process-analysis study

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Abstract

This project presents an Aspen Plus V14 simulation study of a binary benzene–toluene distillation system, with emphasis on process capacity expansion and debottlenecking. A converged base case was first established for a 1000 kg/h feed using a 20-stage RadFrac column, total condenser, kettle reboiler and reflux ratio of 2.5. The model was subsequently tested at 20% and 30% higher throughput while maintaining the same column configuration and reflux ratio. Feed and distillate rates were scaled proportionally, giving 1200/600 kg/h and 1300/650 kg/h respectively. The expanded cases maintained essentially the same thermodynamic separation behavior, while condenser and reboiler duties increased with throughput.

A further operating debottlenecking study was performed at the 30% expanded throughput by increasing the reflux ratio from 2.50 to 2.75. This reduced benzene loss to the bottoms from approximately 0.0984% of the benzene feed to 0.0771%, while increasing condenser duty from 245.52 to 263.06 kW and reboiler duty from 262.21 to 279.75 kW. The study therefore demonstrates the engineering trade-off between separation performance and utility demand. Because no actual equipment hydraulic limit or utility-capacity constraint was specified, the study treats reflux adjustment as an evaluated operating debottlenecking strategy rather than claiming that a physical flooding or equipment bottleneck was proven.

Keywords

Aspen Plus, RadFrac, distillation, benzene, toluene, capacity expansion, debottlenecking, reflux ratio, utility analysis

1. Introduction

Distillation is one of the most widely used separation operations in the chemical process industry. Its performance depends on vapor–liquid equilibrium, number of equilibrium stages, reflux ratio, feed condition, pressure and throughput. When production capacity is increased, a process that operates satisfactorily at its original throughput may experience increased thermal, hydraulic or utility loads.

Process simulation provides a practical method for evaluating these effects before physical modification of a plant. Aspen Plus can be used to establish a material and energy balance, predict column temperature and composition profiles, compare operating cases and evaluate changes in energy demand.

In this study, a binary benzene–toluene distillation system was selected as a compact case study for process debottlenecking. The work focuses on a base case, controlled capacity expansion and an operating modification using reflux-ratio adjustment.

1.1 Problem Statement

The objective is to determine how a distillation column responds to increased production throughput and to evaluate an operating modification that improves separation at the expanded throughput. The analysis considers material balance, product composition, column temperatures and thermal utility requirements.

1.2 Objectives

- Develop and converge a benzene–toluene distillation model in Aspen Plus V14.
- Establish a base case at 1000 kg/h feed throughput.
- Evaluate 20% and 30% proportional capacity expansion cases.
- Compare product flow, separation performance and thermal duties across the cases.
- Evaluate reflux-ratio adjustment as an operating debottlenecking strategy at 30% expansion.
- Quantify the separation improvement and additional condenser/reboiler duty.
- Identify the principal engineering trade-off and define the scope and limitations of the study.

2. Process Description

The simulated process consists of a feed stream containing benzene and toluene. The feed is introduced to a RadFrac distillation column after the upstream pressure/conditioning section represented in the Aspen Plus flowsheet. The column produces a benzene-rich overhead distillate and a toluene-rich bottoms product.

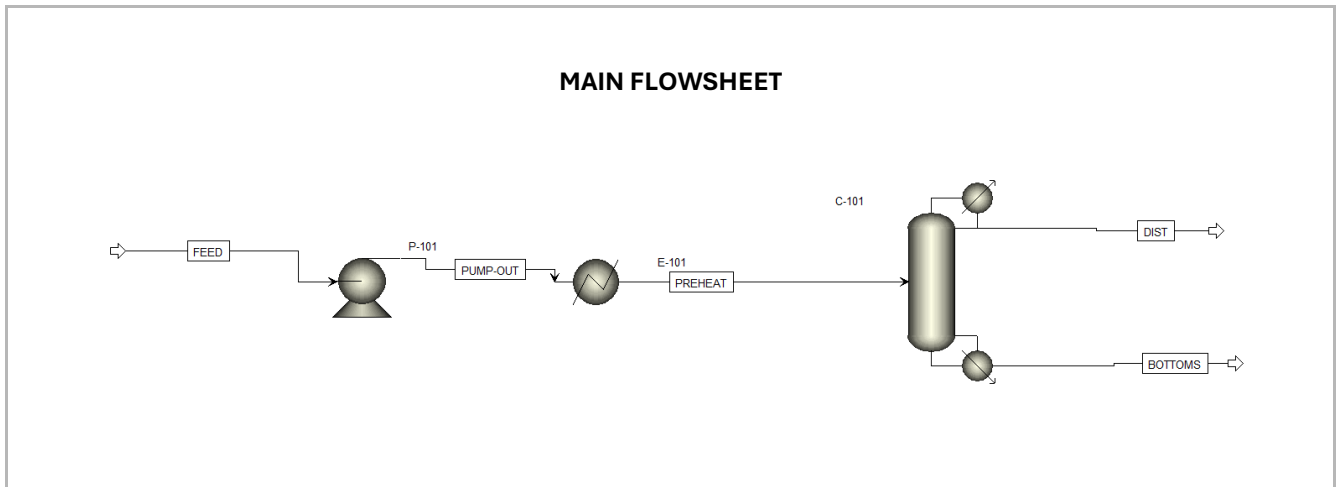


Figure 1. Aspen Plus process flowsheet used for the study.

2.1 Column Configuration

Parameter	Specification
Column	C-101, RadFrac
Number of stages	20 equilibrium stages
Condenser	Total condenser
Reboiler	Kettle reboiler
Base reflux ratio	2.5
Debottlenecked reflux ratio	2.75
Feed temperature	25 °C
Feed pressure	1.5 bar
Column top pressure	1.2 bar
Column bottom pressure	1.295 bar
Feed composition	50 mol% benzene, 50 mol% toluene

3. Simulation Methodology

The simulation was developed in Aspen Plus V14 using a RadFrac equilibrium-stage model. The base case was first converged before changing throughput. Capacity expansion cases were then created by increasing feed and distillate rates proportionally, while holding the column configuration and reflux ratio constant. Finally, the 30% expansion case was copied to a separate debottlenecking case and the reflux ratio was increased from 2.5 to 2.75.

- Define benzene and toluene as the process components.
- Specify a 50/50 molar feed at 25 °C and 1.5 bar.
- Configure C-101 as a 20-stage RadFrac column with total condenser and kettle reboiler.
- Set base distillate rate to 500 kg/h and reflux ratio to 2.5.
- Converge and record base-case material and energy results.
- Increase throughput to 1200 and 1300 kg/h for the 20% and 30% cases, respectively.
- Maintain a 50% mass split between distillate and bottoms for the scaled production cases.
- At 1300 kg/h, increase reflux ratio to 2.75 and record the resulting separation and utility requirements.

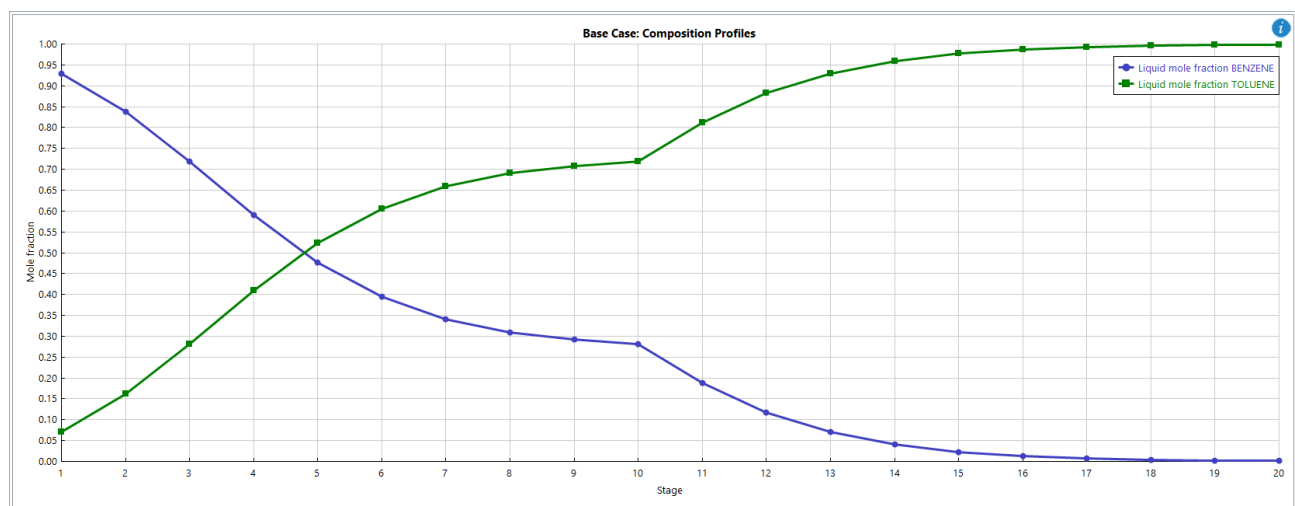
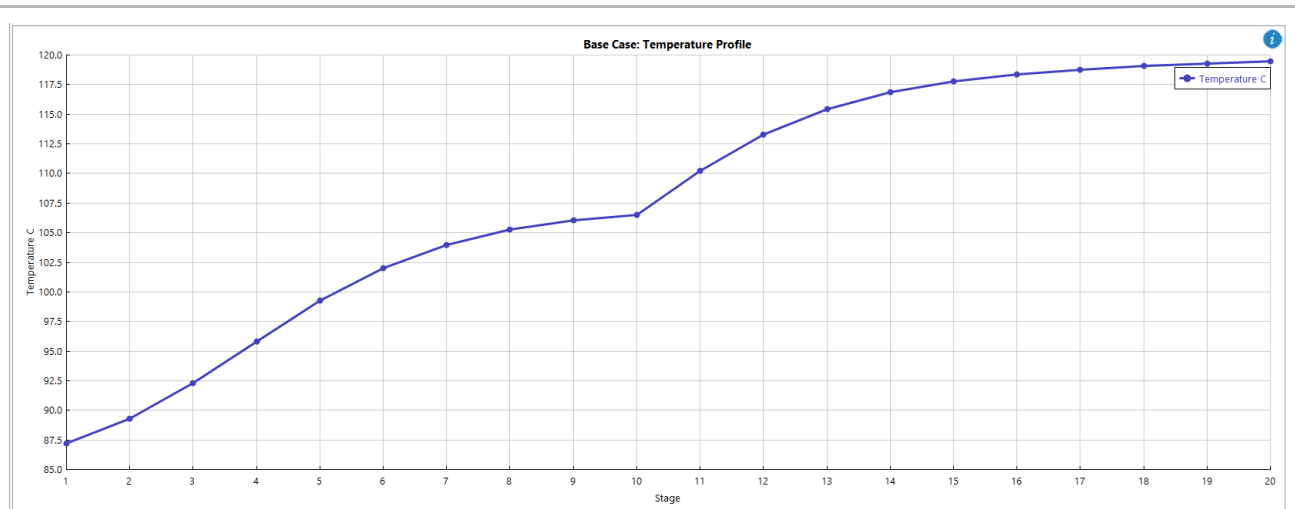
3.1 Case Definition

Case	Feed	Distillate	Bottoms	Reflux Ratio
Base Case	1000 kg/h	500 kg/h	500 kg/h	2.50
20% Expansion	1200 kg/h	600 kg/h	600 kg/h	2.50
30% Expansion	1300 kg/h	650 kg/h	650 kg/h	2.50
Debottlenecked	1300 kg/h	650 kg/h	650 kg/h	2.75

4. Base Case Results

The base case represents the reference operating condition for the expansion study. The 1000 kg/h feed is split into 500 kg/h distillate and 500 kg/h bottoms. The simulated top and bottom temperatures were approximately 87.2 °C and 119.5 °C, respectively.

Parameter	Base Case
Feed rate	1000 kg/h
Distillate rate	500 kg/h
Bottoms rate	500 kg/h
Reflux ratio	2.50
Distillate molar flow	6.31977 kmol/h
Bottoms molar flow	5.42737 kmol/h
Top temperature	87.2065 °C
Bottom temperature	119.489 °C
Condenser duty	45,139.9 cal/s $\approx$ 188.87 kW
Reboiler duty	48,208.3 cal/s $\approx$ 201.70 kW
Benzene recovery to distillate	$\approx$ 99.90%
Distillate benzene purity	$\approx$ 91.67 wt%



5. Capacity Expansion Analysis

Two expansion scenarios were evaluated. The 20% case increases the feed from 1000 to 1200 kg/h and distillate production from 500 to 600 kg/h. The 30% case increases feed to 1300 kg/h and distillate production to 650 kg/h. Column stage count, pressures and reflux ratio were kept unchanged at 20 stages, the specified pressure profile and RR = 2.5.

Parameter	Base	+20%	+30%
Feed (kg/h)	1000	1200	1300
Distillate (kg/h)	500	600	650
Reflux ratio	2.50	2.50	2.50
Condenser duty (kW)	188.87	226.64	245.52
Reboiler duty (kW)	201.70	242.04	262.21
Top temperature (°C)	87.21	87.21	87.21
Bottom temperature (°C)	119.49	119.49	119.49
Benzene recovery	≈99.90%	≈99.90%	≈99.90%
Distillate benzene purity	≈91.67 wt%	≈91.67 wt%	≈91.67 wt%

The expansion cases show that the equilibrium-stage separation profile remains essentially unchanged when throughput is scaled proportionally under the same operating conditions. However, the thermal duties increase substantially. From the base case to 30% expansion, condenser duty increases by approximately 30.0% and reboiler duty increases by approximately 30.0%.

The temperature and composition profiles for the base, 20% expansion and 30% expansion cases remain nearly coincident because the column configuration, feed composition, pressure profile and reflux ratio were maintained constant. The principal effect of throughput expansion is therefore observed in material flow rates and thermal utility requirements rather than in the equilibrium-stage profiles.

6. Debottlenecking Strategy

At the 30% expanded throughput, the reflux ratio was increased from 2.50 to 2.75 while keeping feed rate, distillate rate, number of stages and pressure conditions unchanged. This is treated as an operating debottlenecking strategy intended to provide additional separation driving force without changing the column shell or stage count.

The modification increases internal liquid reflux and therefore increases the thermal load on the condenser and reboiler. The resulting performance was compared against the 30% expansion case at RR = 2.5.

Parameter	30% Expansion	Debottlenecked	Change
Feed	1300 kg/h	1300 kg/h	0%
Distillate	650 kg/h	650 kg/h	0%
Reflux ratio	2.50	2.75	+10.0%
Benzene to bottoms	0.0984%	0.0771%	-21.6%
Benzene recovery	≈99.90%	99.9229%	+0.02 percentage points
Distillate benzene purity	≈91.67 wt%	91.7041 wt%	slight increase
Condenser duty	245.52 kW	263.06 kW	+7.1%
Reboiler duty	262.21 kW	279.75 kW	+6.7%

Split Fraction:

Component	Distillate	Bottoms
Benzene	0.999229039	0.000770961069
Toluene	0.0767709655	0.923229034

7. Results and Discussion

7.1 Overall Performance

The simulation indicates that the 20-stage column can be evaluated at 20% and 30% higher throughput without changing the specified stage count or reflux ratio. The nearly constant temperature endpoints and separation behavior are consistent with the unchanged thermodynamic operating conditions.

The most significant scaling effect is the increase in thermal duty. At 30% expansion, the condenser and reboiler loads rise from approximately 188.87 and 201.70 kW to 245.52 and 262.21 kW, respectively. This makes utility availability an important consideration for any real plant capacity-expansion decision.

7.2 Effect of Reflux Adjustment

Increasing reflux from 2.50 to 2.75 at 1300 kg/h reduces the benzene fraction sent to the bottoms from approximately 0.0984% to 0.0771%. The improvement is achieved with approximately 7.1% higher condenser duty and 6.7% higher reboiler duty relative to the 30% expansion case.

The result illustrates a common process-design trade-off: increased reflux can improve separation, but it also increases internal circulation and therefore energy consumption. The appropriate operating point in an industrial plant would depend on product specifications, utility costs, equipment limits and the economic value of recovered product.

7.3 Utility Scaling

Case	Condenser (kW)	Reboiler (kW)	Total Thermal Duty (kW)
Base	188.87	201.70	390.57
+20%	226.64	242.04	468.68
+30%	245.52	262.21	507.73
Debottlenecked	263.06	279.75	542.81

Relative to the base case, the total of condenser and reboiler duties increases by approximately 30.0% at 30% expansion and by approximately 39.0% in the debottlenecked operating case.

8. Economic Analysis Framework

A fully monetized economic analysis requires site-specific utility prices, annual operating hours, equipment costs and product values. These inputs were not specified for this simulation, so no unsupported monetary values are introduced. Instead, the simulation provides the key engineering quantities required for a subsequent economic evaluation.

Economic Item	Required Input	How this study contributes
Steam/reboiler cost	Steam price, operating hours	Provides reboiler duty
Cooling cost	Cooling-water/refrigeration price, operating hours	Provides condenser duty
Product value	Benzene-rich product value	Provides additional throughput and recovery
Capital cost	Equipment/utility upgrade quotations	Determined only after actual plant constraints
Annual benefit	Product value minus incremental operating cost	Can be calculated from simulation results

For an industrial assessment, the annual incremental utility cost can be estimated from the difference in duty between the selected operating case and the base case, multiplied by annual operating hours and the applicable utility price. The recovered product value can similarly be estimated from incremental product recovery and product price.

9. Limitations and Engineering Scope

- The model is a steady-state equilibrium-stage simulation of a binary benzene–toluene system.
- No actual plant equipment datasheets, utility-capacity limits or operating cost data were provided.
- Column internals and hydraulic calculations were not defined in the model; therefore flooding, weeping, tray loading and hydraulic capacity were not claimed.
- The study does not prove a physical equipment bottleneck. The reflux-ratio change is presented as an evaluated operating debottlenecking strategy.
- Economic conclusions are therefore qualitative/framework-based unless site-specific utility and capital-cost data are supplied.
- The results are intended for process-analysis and screening purposes rather than detailed plant modification or final equipment design.

10. Conclusion

A benzene–toluene distillation process was successfully modeled and converged in Aspen Plus V14 using a 20-stage RadFrac column with total condenser and kettle reboiler. The base case processed 1000 kg/h of feed and produced 500 kg/h of distillate at a reflux ratio of 2.5.

Capacity expansion simulations were performed at 1200 and 1300 kg/h feed throughput, corresponding to 20% and 30% expansion. The expanded cases maintained essentially the same separation behavior under unchanged operating conditions, while condenser and reboiler duties increased with throughput.

At the 30% expanded throughput, increasing reflux ratio from 2.5 to 2.75 reduced benzene loss to the bottoms by approximately 21.6% and increased benzene recovery to approximately 99.923%. The improvement required approximately 7.1% additional condenser duty and 6.7% additional reboiler duty compared with the 30% expansion case.

Overall, the study demonstrates how Aspen Plus can be used to evaluate capacity expansion, separation performance and utility trade-offs before considering physical plant modifications. For a real industrial debottlenecking study, the next engineering step would be to compare these thermal loads with actual condenser/reboiler capacities and hydraulic limits and then complete a site-specific economic assessment.

11. References

1. Aspen Technology, Aspen Plus V14 Documentation and User Guidance, AspenTech.
2. Wankat, P. C., Separation Process Engineering, Prentice Hall.
3. Seader, J. D., Henley, E. J. and Roper, D. K., Separation Process Principles, Wiley.
4. Kister, H. Z., Distillation Design, McGraw-Hill.

Appendix A — Aspen Plus Results to Insert

To make the final submitted report presentation-ready, replace the placeholders in the report with the corresponding screenshots already generated during the Aspen Plus study.

Figure	Recommended screenshot
Fig. 1	Final main flowsheet
Fig. 2	Base temperature profile
Fig. 3	Base composition profile
Fig. 4	20% expansion temperature/composition profile
Fig. 5	30% expansion temperature/composition profile
Fig. 6	Debottlenecked C-101 Summary
Fig. 7	Debottlenecked Split Fraction
Optional	Pressure, liquid/vapor flow and enthalpy profiles if space permits

Appendix B — Key Aspen Inputs for Reproduction

Input	Value
Components	BENZENE, TOLUENE
Feed composition	0.50 benzene / 0.50 toluene, mol fraction
Feed temperature	25 °C
Feed pressure	1.5 bar
Column stages	20
Condenser	Total
Reboiler	Kettle
Column top pressure	1.2 bar
Column bottom pressure	1.295 bar
Base/expansion RR	2.5
Debottleneck RR	2.75
Base feed/distillate	1000 / 500 kg/h
20% feed/distillate	1200 / 600 kg/h
30% feed/distillate	1300 / 650 kg/h